Analytical Chemistry Cumulative Exam
February 13, 2026

This exam focuses on basic undergraduate analytical chemistry: equilibria, titrations and simple instrumental analysis involving principles of quantitative analysis.
(At end of exam are equations that might or might not be helpful in answering these questions)

1. **(5 points)** What is the pH of an aqueous solution containing **1.0×10^{-9} M HCl?** *(show work)*

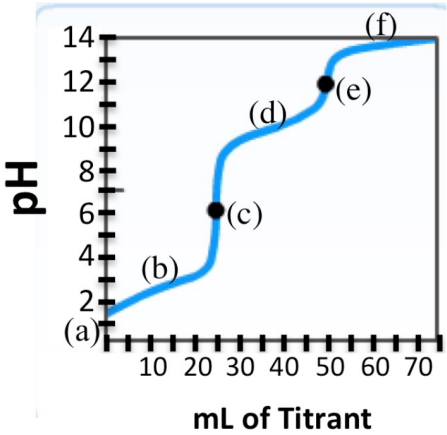
2. **(18 points total)** A weak acid, H_2A, (0.100 M) is titrated with a strong base. Use the figure here to answer the following questions. (The y-axis ranges from pH = 0 to 14.)

(a) (5 points) What ***letter*** in the figure best represents the location of the second equivalence point? Explain why?

(b) (3 points) Estimate the ***value*** of pK_{a1} (not the letter) from the figure where $[H_2A] / [HA^-] = 1$.

(c) (5 points) Estimate the ***value*** of K_{a2} from the figure. *(Show work.)*

(d) (5 points) Suggest a suitable indicator (from the table below) that will change color at the equivalence point when the titration has converted all the H_2A to A^{2-}. Explain._____



Acid/Base Indicators

Common Name	*Transition Range, pH*	*pKa**	*Color Change*
Thymol blue	1.2-2.8	1.65	R-Y
	8.0-9.6	8.96	Y-B
Methyl yellow	2.9-4.0		R-Y
Bromocresol green	3.8-5.4	4.66	Y-B
Bromocresol purple	5.2-6.8	6.12	Y-P
Bromothymol blue	6.2-7.6	7.10	Y-B
Phenol red	6.8-8.4	7.81	Y-R
Phenolphthalein	8.3-10.0		C-R
Thymolphthalein	9.3-10.5		C-B
Nitramine	10.8-13		C-O/Br

(C = colorless, R = red, Y = yellow, B = blue, P = purple, O = orange, Br = Brown)

3. (21 points) A solution of **25.00 mL** of **0.1000 M** disodium maleate NaOOC-CH=CH-COONa (the deprotonated form of maleate, which can be represented by A^{2-}) is titrated with a strong acid. Note that the two K_a's of the diprotic form of maleate (maleic acid) are: $\mathbf{K_{a1} = 1.3 \times 10^{-2}}$, $\mathbf{K_{a2} = 5.9 \times 10^{-7}}$. ***Answer all three of the following questions for this titration.***

(a) **(7 points)** What is the pH of the solution before the titration begins (**0.1000 M of A^{2-}**)? *(show work)*

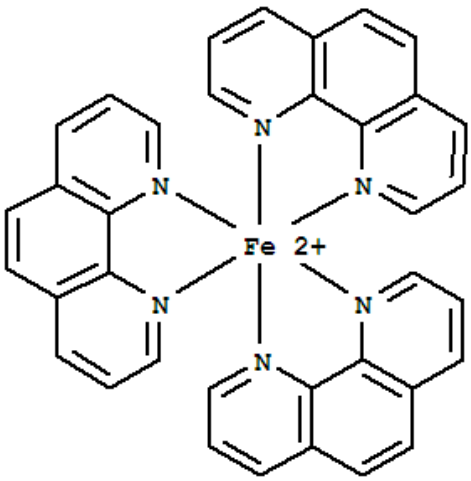
(b) **(7 points)** What is the pH of the solution ***after*** adding **50.00 mL** of **0.0500 M HCl**? *(show work)*

(c) **(7 points)** What is the pH of the solution ***after*** adding **150.00 mL** of **0.0500 M HCl**? *(show work)*

4. (12 points) A 20.0 mL sample of 0.0100 M Cu^{2+} buffered at pH 10.00 is titrated with 0.0200 M EDTA. Calculate pCu^{2+} when a volume of 10.0 mL EDTA has been added. The formation equilibrium constant, K_f, for the Cu^{2+}-EDTA complex is**: $\mathbf{\log(K_f) = 18.78}$**. The fraction of free EDTA in the Y^{4-} form, α_4 or α_{Y4-}, can be found in the table in the equation sheet. *(show work; assume 25 °C).*

5. (20 points) Calculate the molar solubility of $PbCO_{3(s)}$ in a solution buffered to a pH of 5.00. (Ignore chemistry involving CO_2). Acid dissociation constants for carbonic acid are: $\mathbf{K_{a1} = 4.46 \times 10^{-7}}$ and $\mathbf{K_{a2} = 4.69 \times 10^{-11}}$. The solubility product for $PbCO_{3(s)}$ is: $\mathbf{K_{sp} = .4 \times 10^{-14}}$. *(show work, state assumptions if necessary)*

6. (24 points) Name two different instrumental methods that can be used to ***quantify*** the compound shown below, Tris(1,10-phenanthroline)iron(II), in a sample. Explain why each method would be appropriate. Note that this analyte is in a mixture with other compounds that arrives to your laboratory in a solid form. Describe the sample preparation procedures that you would perform, if any, before the instrumental analysis. Note any assumptions that you would have to make, if any, in your analysis approach.



http://www.guidechem.com/reference/dic-329192.html

Equations relevant to acid-base equilibria

$$[H_3O^+]^3 + K_a\ [H_3O^+]^2 - \left(K_w + K_a\ C_{HA}\right)\ [H_3O^+] - K_a\ K_w = 0$$

$$[H_3O^+] \approx \frac{-K_a + \sqrt{K_a^2 + 4\ K_a\ C_{HA}}}{2}$$

$$[H_3O^+] \approx \sqrt{K_a\ C_{HA}}$$

$$\mathbf{[OH^-]^3 + K_b\ [OH^-]^2 - \left(K_w + K_b\ C_{NaA}\right)\ [OH^-] - K_b\ K_w = 0}$$

$$\mathbf{[OH^-] \approx \frac{-K_b + \sqrt{K_b^2 + 4\ K_b\ C_{NaA}}}{2}}$$

$$\mathbf{[OH^-] \approx \sqrt{K_b\ C_{NaA}}}$$

$$\mathbf{[H_3O^+] = \sqrt{\frac{K_{a_2}\ [HA^-] + K_w}{1 + [HA^-]/K_{a_1}}}}$$

$$\mathbf{[H_3O^+] = \sqrt{\frac{K_{a_2}\ C_{NaHA} + K_w}{1 + C_{NaHA}/K_{a_1}}}}$$

$$\mathbf{[H_3O^+] \approx \sqrt{K_{a_1}\ K_{a_2}}}$$

$$pH = -\log[H_3O^+] \approx pKa + \log\left(\frac{C_{NaA}}{C_{HA}}\right)$$

Equations Relevant to EDTA Equilibria

$$K'_{MY} = \alpha_4 K_{MY} = \frac{[MY^{n-4}]}{[M^{n+}]\, C_T}$$

Free EDTA as Y4–

Values of $\alpha_{Y^{4-}}$ for EDTA at 25°C and μ=0.10M

pH	α_{Y4-}
0	1.3×10^{-23}
1	1.4×10^{-18}
2	2.6×10^{-14}
3	2.1×10^{-11}
4	3.0×10^{-9}
5	2.9×10^{-7}
6	1.8×10^{-5}
7	3.8×10^{-4}
8	4.2×10^{-3}
9	0.041
10	0.30
11	0.81
12	0.98
13	1.00
14	1.00

Reference: *Quantitative Chemical Analysis, 8th Edition, Harris*